Barrier Certificates for an Asymmetric Indefinite Riccati Equation in a Partially Observed LQ Mean-Field Game

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Abstract

Chen, Nie, and Wu derive a feedback representation for an indefinite partially observed linear-quadratic mean-field game, conditional on solvability of an asymmetric indefinite Riccati equation. Their paper explicitly notes that the general solvability of this asymmetric equation remains open. This note gives partial progress, not a complete solution of that open problem. We prove two rigorous sufficient conditions. The first is an a posteriori centered barrier around a chosen reference path. The second is a simpler zero-centered norm barrier leading to scalar and constant small-gain tests. Under either certificate, the Riccati terminal-value problem has a bounded absolutely continuous solution, the effective control matrix is uniformly positive along the solution, and the solution is unique in the natural uniformly positive feedback class.

1 Status and relation to the source paper

The equation studied below is the asymmetric Riccati equation appearing as equation (35) in Chen–Nie–Wu [1]. In their Theorem 4.12, the feedback representation is derived after assuming that this equation admits a unique solution. Their Remark 4.13(ii) then states that the solvability of this asymmetric indefinite Riccati equation is still open.

The results below are therefore best understood as *partial progress*. They do not characterize solvability and do not solve the full indefinite-control-weight problem. They give verifiable certificates: once the stated barrier inequalities are checked, the missing Riccati solvability assumption used in the feedback construction is valid in the certified class.

2 The Riccati equation and notation

All coefficient functions are deterministic and essentially bounded on $[0,T]$. We use the dimensions of [1]: $A, \bar{A}, D, \bar{D}, Q, L_T \in \mathbb{R}^{n\times n}$, $B, \bar{B}, F, \bar{F}, S \in \mathbb{R}^{n\times m}$, and $R \in \mathbb{R}^{m\times m}$. The spectral norm is denoted by $\|\cdot\|$, and

$$\operatorname{Sym} M := \frac{M + M^\top}{2}.$$

Set

$$\widehat{A}_t := A_t + \bar{A}_t, \qquad \widehat{B}_t := B_t + \bar{B}_t, \qquad \widehat{D}_t := D_t + \bar{D}_t, \qquad \widehat{F}_t := F_t + \bar{F}_t,$$

and put

$$\Xi := (1-\alpha_3)L_T.$$

For $P \in \mathbb{R}^{n\times n}$ define

$$\mathcal{R}_t(P) := (1-\beta_1)R_t + F_t^\top P \widehat{F}_t, \tag{1}$$

$$\mathcal{U}_t(P) := P\widehat{B}_t + D_t^\top P \widehat{F}_t + (1-\beta_2)S_t, \tag{2}$$

$$\mathcal{V}_t(P) := PB_t + \widehat{D}_t^\top P F_t + (1-\alpha_2)S_t. \tag{3}$$

The Riccati equation is

$$\begin{cases} 0 = \dot{\Sigma}_t + \Sigma_t \widehat{A}_t + A_t^\top \Sigma_t + D_t^\top \Sigma_t \widehat{D}_t - \mathcal{U}_t(\Sigma_t)\mathcal{R}_t(\Sigma_t)^{-1}\mathcal{V}_t(\Sigma_t)^\top + (1-\alpha_1)Q_t, \\ \Sigma_T = \Xi. \end{cases} \tag{4}$$

Define the vector field, where the inverse exists, by

$$\Phi_t(P) := P\widehat{A}_t + A_t^\top P + D_t^\top P \widehat{D}_t - \mathcal{U}_t(P)\mathcal{R}_t(P)^{-1}\mathcal{V}_t(P)^\top + (1-\alpha_1)Q_t. \tag{5}$$

Then (4) is equivalent to the backward integral equation

$$\Sigma_t = \Xi + \int_t^T \Phi_s(\Sigma_s)\,\mathrm{d}s. \tag{6}$$

The equation is generally asymmetric: even when the data Q, L_T are symmetric, the two factors $\mathcal{U}_t(\Sigma_t)$ and $\mathcal{V}_t(\Sigma_t)$ need not coincide.

For later estimates introduce

$$\begin{aligned} a_t &:= \left\|\widehat{A}_t\right\| + \|A_t\| + \|D_t\| \left\|\widehat{D}_t\right\|, & u_t &:= \left\|\widehat{B}_t\right\| + \|D_t\| \left\|\widehat{F}_t\right\|, \\ v_t &:= \|B_t\| + \left\|\widehat{D}_t\right\| \|F_t\|, & w_t &:= \|F_t\| \left\|\widehat{F}_t\right\|, \\ s_t^u &:= |1-\beta_2| \, \|S_t\|, & s_t^v &:= |1-\alpha_2| \, \|S_t\|, \\ q_t &:= |1-\alpha_1| \, \|Q_t\|, & M_0 &:= \|\Xi\| = |1-\alpha_3| \, \|L_T\|. \end{aligned}$$

For $K \geq 0$ and $r > 0$ define the local Lipschitz majorant

$$L_t(K,r) := a_t + \frac{u_t(v_t K + s_t^v) + (u_t K + s_t^u)v_t}{r} + \frac{(u_t K + s_t^u)(v_t K + s_t^v)w_t}{r^2}. \tag{7}$$

3 Two elementary estimates

Lemma 3.1 (Symmetric-part positivity implies inverse control). *If $M \in \mathbb{R}^{m\times m}$ and $\operatorname{Sym} M \succeq rI_m$ for some $r > 0$, then M is invertible and $\left\|M^{-1}\right\| \leq r^{-1}$.*

Proof. For every $x \in \mathbb{R}^m$,

$$r\,\|x\|^2 \leq x^\top(\operatorname{Sym} M)x = x^\top M x \leq |x^\top M x| \leq \|Mx\| \, \|x\|.$$

Thus $\|Mx\| \geq r\,\|x\|$ for all x, so the least singular value of M is at least r. □

Lemma 3.2 (Local Lipschitz bound). *Fix t outside a null set. Suppose $\|P\|, \|P'\| \leq K$ and*

$$\operatorname{Sym} \mathcal{R}_t(P) \succeq rI_m, \qquad \operatorname{Sym} \mathcal{R}_t(P') \succeq rI_m.$$

Then

$$\left\|\Phi_t(P) - \Phi_t(P')\right\| \leq L_t(K,r) \left\|P - P'\right\|. \tag{8}$$

Proof. The linear part of Φ contributes at most $a_t \|P - P'\|$. Moreover,

$$\begin{aligned} \left\|\mathcal{U}_t(P) - \mathcal{U}_t(P')\right\| &\leq u_t \left\|P - P'\right\|, \\ \left\|\mathcal{V}_t(P) - \mathcal{V}_t(P')\right\| &\leq v_t \left\|P - P'\right\|, \\ \left\|\mathcal{R}_t(P) - \mathcal{R}_t(P')\right\| &\leq w_t \left\|P - P'\right\|. \end{aligned}$$

On the ball $\|P\| \leq K$ we also have

$$\|\mathcal{U}_t(P)\| \leq u_t K + s_t^u, \qquad \|\mathcal{V}_t(P)\| \leq v_t K + s_t^v.$$

By Theorem 3.1, both inverses have norm at most r^{-1}. Expanding the difference of the fractional terms and using

$$R^{-1} - (R')^{-1} = R^{-1}(R' - R)(R')^{-1}$$

gives exactly (8) with $L_t(K,r)$ as in (7). □

4 A centered a posteriori barrier

The first certificate is centered around a reference path. This is more flexible than a zero-centered norm estimate because the tube need not contain the zero matrix.

Definition 4.1 (Centered Riccati barrier)**.** Let $C \in W^{1,\infty}([0,T];\mathbb{R}^{n\times n})$, let $h : [0,T] \to [0,\infty)$ be continuous, and let $r : [0,T] \to (0,\infty)$ be measurable with $\operatorname{ess\,inf}_{[0,T]} r > 0$. Put $K_t := \|C_t\| + h_t$. The triple (C, h, r) is a centered Riccati barrier if the following conditions hold.

First, the effective control matrix is uniformly positive on the tube around C:

$$\operatorname{Sym} \mathcal{R}_t(P) \succeq r_t I_m \quad \text{for a.e. } t \in [0,T] \quad \text{and all } P \text{ with } \|P - C_t\| \le h_t. \tag{9}$$

Second, the radius dominates the residual and the local Lipschitz growth:

$$h_t \ge \|\Xi - C_T\| + \int_t^T \Big[\Big\|\dot{C}_s + \Phi_s(C_s)\Big\| + L_s(K_s, r_s) h_s\Big]\, \mathrm{d}s, \qquad 0 \le t \le T. \tag{10}$$

Theorem 4.2 (Centered barrier criterion)**.** *If a centered Riccati barrier (C, h, r) exists, then (4) has a solution*

$$\Sigma \in W^{1,\infty}([0,T];\mathbb{R}^{n\times n})$$

satisfying

$$\|\Sigma_t - C_t\| \le h_t \quad (0 \le t \le T), \qquad \operatorname{Sym} \mathcal{R}_t(\Sigma_t) \succeq r_t I_m \quad \textit{for a.e. } t. \tag{11}$$

This solution is unique among all absolutely continuous solutions Y of (4) satisfying

$$\operatorname{Sym} \mathcal{R}_t(Y_t) \succeq \eta_Y I_m \quad \textit{for a.e. } t \in [0,T] \tag{12}$$

for some constant $\eta_Y > 0$.

Proof. Let

$$\mathcal{B} := \{Z \in C([0,T];\mathbb{R}^{n\times n}) : \|Z_t\| \le h_t \text{ for all } t \in [0,T]\}.$$

For $Z \in \mathcal{B}$ define

$$(\mathcal{T}Z)_t := \Xi - C_t + \int_t^T \Phi_s(C_s + Z_s)\, \mathrm{d}s.$$

Since $C_t = C_T - \int_t^T \dot{C}_s\, \mathrm{d}s$, this can be written as

$$(\mathcal{T}Z)_t = \Xi - C_T + \int_t^T \big[\dot{C}_s + \Phi_s(C_s + Z_s)\big]\, \mathrm{d}s.$$

By (9), Theorem 3.1, and Theorem 3.2, for a.e. s,

$$\|\Phi_s(C_s + Z_s) - \Phi_s(C_s)\| \le L_s(K_s, r_s) h_s.$$

Therefore (10) implies $\|(\mathcal{T}Z)_t\| \le h_t$ for every t, so $\mathcal{T}$ maps $\mathcal{B}$ into itself.

The image $\mathcal{T}(\mathcal{B})$ is uniformly bounded and equicontinuous, because all terms in the preceding bound are integrable. The map $P \mapsto \Phi_t(P)$ is continuous on the tube for a.e. t and is dominated there by an integrable function. Dominated convergence gives continuity of $\mathcal{T}$ on $\mathcal{B}$, and Arzela-Ascoli gives compactness of $\mathcal{T}(\mathcal{B})$. Schauder's fixed-point theorem yields $Z \in \mathcal{B}$ with $\mathcal{T}Z = Z$. Then $\Sigma := C + Z$ satisfies (6), hence (4), and (11) follows. The solution belongs to $W^{1,\infty}$ because the vector field is essentially bounded on the certified tube.

It remains to prove uniqueness in the uniformly positive class. Let Y be another solution satisfying (12). Both Y and Σ are continuous on a compact interval, hence bounded. Also

$\mathcal{R}_t(Y_t)^{-1}$ and $\mathcal{R}_t(\Sigma_t)^{-1}$ are essentially bounded by Theorem 3.1. Repeating the algebra in Theorem 3.2 along the two paths gives some $L^Y \in L^1(0,T)$ such that

$$\|\Phi_t(Y_t) - \Phi_t(\Sigma_t)\| \le L_t^Y \|Y_t - \Sigma_t\| \quad \text{for a.e. } t.$$

Since $Y_T = \Sigma_T = \Xi$,

$$\|Y_t - \Sigma_t\| \le \int_t^T L_s^Y \|Y_s - \Sigma_s\| \, \mathrm{d}s.$$

The backward Gronwall inequality gives $Y = \Sigma$. □

5 A zero-centered norm barrier

The centered theorem is flexible but requires a reference path. The following direct criterion is less flexible but easier to check.

Definition 5.1 (Zero-centered Riccati barrier)**.** A pair (k, r) is a zero-centered Riccati barrier if $k : [0,T] \to [0,\infty)$ is continuous, $r : [0,T] \to (0,\infty)$ is measurable, $\operatorname{ess\,inf}_{[0,T]} r > 0$, and

$$\operatorname{Sym} \mathcal{R}_t(P) \succeq r_t I_m \quad \text{for a.e. } t \in [0,T] \quad \text{and all } P \text{ with } \|P\| \le k_t, \tag{13}$$

while

$$k_t \ge M_0 + \int_t^T \left[a_s k_s + q_s + \frac{(u_s k_s + s_s^u)(v_s k_s + s_s^v)}{r_s} \right] \mathrm{d}s \quad (0 \le t \le T). \tag{14}$$

Theorem 5.2 (Zero-centered criterion)**.** *If a zero-centered Riccati barrier (k, r) exists, then (4) has a solution $\Sigma \in W^{1,\infty}$ satisfying*

$$\|\Sigma_t\| \le k_t, \qquad \operatorname{Sym} \mathcal{R}_t(\Sigma_t) \succeq r_t I_m \quad \textit{for a.e. } t.$$

The solution is unique in the uniformly positive class (12).

Proof. Let $\mathcal{B} := \{X \in C([0,T]; \mathbb{R}^{n\times n}) : \|X_t\| \le k_t \text{ for all } t\}$. For $X \in \mathcal{B}$, condition (13) and Theorem 3.1 give $\left\|\mathcal{R}_t(X_t)^{-1}\right\| \le r_t^{-1}$. Hence

$$\|\Phi_t(X_t)\| \le a_t k_t + q_t + \frac{(u_t k_t + s_t^u)(v_t k_t + s_t^v)}{r_t} \quad \text{for a.e. } t.$$

The integral operator $(\mathcal{T}X)_t := \Xi + \int_t^T \Phi_s(X_s)\,\mathrm{d}s$ maps $\mathcal{B}$ into itself by (14). As in the proof of Theorem 4.2, this operator is continuous and compact, so Schauder's theorem gives a fixed point. This fixed point is the desired solution. Uniqueness in the uniformly positive class is exactly the uniqueness argument at the end of the proof of Theorem 4.2. □

6 Scalar and constant tests

The robust tube condition can be checked directly. A simple sufficient lower bound is as follows. Define

$$\rho_t := \lambda_{\min}(\operatorname{Sym}((1-\beta_1)R_t)), \qquad w_t := \|F_t\| \left\|\widehat{F}_t\right\|. \tag{15}$$

If $\|P\| \le k_t$, then

$$\operatorname{Sym} \mathcal{R}_t(P) \succeq (\rho_t - w_t k_t) I_m. \tag{16}$$

Indeed, $\lambda_{\min}(\operatorname{Sym}(F_t^\top P \widehat{F}_t)) \ge -\left\|F_t^\top P \widehat{F}_t\right\| \ge -w_t \|P\|$.

Corollary 6.1 (Scalar majorant test)**.** *Suppose there are a continuous $k : [0,T] \to [0,\infty)$ and $\varepsilon > 0$ such that*

$$\rho_t - w_t k_t \ge \varepsilon \quad \textit{for a.e. } t \in [0,T], \tag{17}$$

and

$$k_t \ge M_0 + \int_t^T \left[a_s k_s + q_s + \frac{(u_s k_s + s_s^u)(v_s k_s + s_s^v)}{\rho_s - w_s k_s} \right] \mathrm{d}s \quad (0 \le t \le T). \tag{18}$$

Then (4) *has a unique solution in the uniformly positive class, and this solution satisfies*

$$\|\Sigma_t\| \le k_t, \qquad \operatorname{Sym} \mathcal{R}_t(\Sigma_t) \succeq (\rho_t - w_t k_t) I_m \succeq \varepsilon I_m$$

for a.e. t.

Proof. Set $r_t := \rho_t - w_t k_t$ and apply Theorem 5.2 using (16). □

Let

$$\begin{gathered} \bar{a} := \operatorname{ess\,sup} a_t, \qquad \bar{u} := \operatorname{ess\,sup} u_t, \qquad \bar{v} := \operatorname{ess\,sup} v_t, \\ \bar{s}^u := \operatorname{ess\,sup} s_t^u, \qquad \bar{s}^v := \operatorname{ess\,sup} s_t^v, \qquad \bar{q} := \operatorname{ess\,sup} q_t, \\ \rho_0 := \operatorname{ess\,inf} \rho_t, \qquad \bar{w} := \operatorname{ess\,sup} w_t. \end{gathered}$$

Corollary 6.2 (Constant small-gain test)**.** *Assume there is a constant $K \ge M_0$ such that*

$$\rho_0 - \bar{w} K > 0 \tag{19}$$

and

$$M_0 + T \left[\bar{a} K + \bar{q} + \frac{(\bar{u} K + \bar{s}^u)(\bar{v} K + \bar{s}^v)}{\rho_0 - \bar{w} K} \right] \le K. \tag{20}$$

Then (4) *has a unique solution in the uniformly positive class, and*

$$\|\Sigma_t\| \le K, \qquad \operatorname{Sym} \mathcal{R}_t(\Sigma_t) \succeq (\rho_0 - \bar{w} K) I_m \quad \textit{for a.e. } t.$$

In particular, if $\rho_0 > \bar{w} M_0$, then the two displayed assumptions hold on all sufficiently short horizons after choosing any K with $M_0 < K < \rho_0 / \bar{w}$, with the upper bound omitted when $\bar{w} = 0$.

Proof. Apply Theorem 6.1 with $k_t \equiv K$. □

7 Consequence for feedback coefficients

The feedback theorem in [1] also uses a symmetric Riccati equation, usually denoted by Π, and a linear terminal-value equation for an affine term. The results above address only the asymmetric Riccati equation (4); they do not replace the assumptions needed for the symmetric Riccati equation.

Assume, however, that the symmetric Riccati part is already controlled as in the source paper, and that either Theorem 4.2 or Theorem 5.2 applies to the asymmetric equation. Then

$$\mathcal{R}_t(\Sigma_t)^{-1} \mathcal{V}_t(\Sigma_t)^\top$$

is essentially bounded. In the zero-centered case, for instance,

$$\left\| \mathcal{R}_t(\Sigma_t)^{-1} \mathcal{V}_t(\Sigma_t)^\top \right\| \le r_t^{-1} (v_t k_t + s_t^v).$$

The centered case is identical with k_t replaced by $\|C_t\| + h_t$. Thus the remaining affine terminal-value equation has bounded coefficients, and the limiting average equation and individual closed-loop equations are linear SDEs with bounded coefficients. This supplies the missing Riccati well-posedness input in the certified regime.

8 Limitations

The criteria are sufficient, not necessary. The zero-centered condition is especially conservative: since the tube $\|P\| \le k_t$ contains $P = 0$, it forces

$$\mathrm{Sym}((1-\beta_1)R_t) \succeq r_t I_m \quad \text{for a.e. } t$$

under the direct barrier hypothesis. Therefore it does not handle cases where the nominal control block is genuinely indefinite and positivity is recovered only through the solution-dependent term $F_t^\top \Sigma_t \widehat{F}_t$.

The centered certificate is less restrictive because the tube may be placed away from zero, but it requires a reference path and an a posteriori verification of positivity and residual control. Consequently, the results here are genuine partial progress on the solvability of equation (35), but not a complete solution of the open problem stated in Remark 4.13(ii) of [1].

References

[1] Tian Chen, Tianyang Nie, and Zhen Wu. *Indefinite Linear-Quadratic Partially Observed Mean-Field Game.* arXiv:2508.01568, 2025. https://arxiv.org/abs/2508.01568.

[2] Eberhard Zeidler. *Nonlinear Functional Analysis and its Applications. I: Fixed-Point Theorems.* Springer, 1986.